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## Assignment 2

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Note: Please include references when applying results from the textbook [Perko].

Problem 1: The Duffing system reads

$$\dot{x}_1 = f_1(x_1, x_2) = x_2, \quad (1)$$

$$\dot{x}_2 = f_2(x_1, x_2) = \mu x_1 - x_1^3. \quad (2)$$

Here the two variables  $x_1$  and  $x_2$  depend on time  $t$ . A dot above a time dependent variable denotes differentiation with respect to  $t$ . The parameter  $\mu$  is a real number.

- Find all critical points of the two dimensional vectorfield  $\mathbf{f} = (f_1, f_2)^T$  in the system (1-2).
- Determine the stability of all critical points for the cases  $\mu > 0$  and  $\mu < 0$ . Classify these critical points according to the list: saddles, centers, cusps, nodes and foci. Hint: Use the theorems of chapter 2.10 in the textbook [Perko].
- Sketch the flow in a phase plane plot for  $\mu > 0$  and  $\mu < 0$ . Use pencil and paper and support your findings using pplane or Maple (hint: you may use the command *plotfield* in Maple).
- For  $\mu > 0$  there are homoclinic orbits. Similar to finding center manifolds we can determine the homoclinic orbits by introducing a function  $h \in C^1$  mapping  $\mathbb{R} \rightarrow \mathbb{R}$  according to

$$x_2 = h(x_1). \quad (3)$$

From the sketch of the phase plane plot it is seen that  $h$  is a one to one mapping in the first quadrant of the  $x_1x_2$ -plane ( $x_1 > 0$  and  $x_2 > 0$ ). Which condition must  $h$  satisfy? Derive a first order differential equation for  $h$  from (1-2) and solve this equation analytically.

- Plot your analytical expression for  $h(x_1)$  together with the vectorfield. Does the figure show what you expect? Discuss in few words your findings.
- Determine an analytical expression for a graph  $h$  of the homoclinic orbit in the 4th quadrant of the  $x_1x_2$ -plane ( $x_1 > 0$  and  $x_2 < 0$ ).

Problem 2: The results from problem 1 indicate that a bifurcation occurs at  $\mu = 0$  and that  $\mu$  is a bifurcation parameter. Secondly, for  $\mu = 0$  the critical point  $(x_1, x_2) = (0, 0)$  is different compared to the cases where  $\mu \neq 0$ .

- Construct a bifurcation diagram for the system in (1-2). In doing so sketch in a figure  $x_{01}$  as function of  $\mu$ , for the critical points  $(x_{01}, x_{02})$ .
- Determine the type of bifurcation you observe.
- For  $\mu = 0$  the system in (1-2) becomes

$$\dot{x}_1 = x_2, \quad (4)$$

$$\dot{x}_2 = -x_1^3. \quad (5)$$

Is the critical point  $(x_1, x_2) = (0, 0)$  hyperbolic or nonhyperbolic for  $\mu = 0$ ?

- d) Determine the stability of the critical point  $(x_1, x_2) = (0, 0)$  by finding a suitable Liapunov function  $V(x_1, x_2)$ .
- e) Make a contour plot of your Liapunov function  $V(x_1, x_2)$  using e.g. Maple. Compare to the vectorfield using pplane. Are the contour lines solution trajectories of (4-5)?
- f) What information about the stability of the critical point  $(0, 0)$  can you get from Theorem 2, p151, in section 2.11 of the textbook by Perko?